Journey Inside the Atom

Everything we see, feel, or touch is made up of matter. The matter consists of tiny particles called atoms. For centuries, scientists have wondered about the hidden world inside the atom.
In this chapter, we will see the fascinating journey of how the concept of the atom developed from ancient ideas to modern models and explore the subatomic particles that make up every atom in the universe.

Rediscovering the Roots of Atomic Theory

- Acharya Kanada, an Indian philosopher, suggested that if matter (dravya) is divided again, one would eventually reach the smallest particles that cannot be divided any further. He called these particles **parmanus**. His ideas are recorded in the Sanskrit text ***Vaisesika Sutras***. A parmanu is infinitely small and cannot be identified by the senses. Combinations of parmanus form the entire material world.
- The Greek philosophers **Leucippus** and **Democritus** proposed a similar idea. They called these indivisible particles **atomos** (Greek for 'indivisible'). It is from this Greek word that the modern term 'atom' is derived.

Note The concept of the atom originally emerged as a philosophical idea, not from scientific experiments. It was later placed on a scientific journey by **John Dalton**.

Dalton's Atomic Theory (1808)

Many centuries later, in 1808, John Dalton proposed his atomic theory based on scientific experiments.

Its main points are

- All matter is composed of extremely small, indivisible particles called atoms.
- Atoms of the same element are identical in mass and properties.
- Atoms of different elements differ in mass and properties.
- Atoms cannot be created, divided, or destroyed in a chemical reaction.
- Atoms combine in simple, whole-number ratios to form compounds.

Dalton's atomic theory was the first scientific description of how matter is made. It became the starting point for the modern understanding of atomic structure.

A Short Historical Journey Through Atomic Models

Following Dalton's atomic theory, scientists became curious: What are atoms made of? What do they look like? How do the atoms of different elements differ from each other? Over the next century, a series of great experiments led to new and modified models of the atom.

Discovery of the Electron: Cathode Rays

- In 1897, **J. J. Thomson** studied the conduction of electric current through gases at very low pressure using a glass tube with two electrodes. He applied a high voltage and observed rays moving from the **cathode** (negative electrode) to the **anode** (positive electrode). These were called **cathode rays**.
- By studying these cathode rays in electric and magnetic fields, Thomson concluded that they are streams of negatively charged particles with a mass much smaller than that of atoms. These particles were later called **electrons**. Since electrons were emitted from atoms, this proved that atoms are made up of smaller subatomic particles called electrons.

Expert Insight

The nature of cathode rays was found to be independent of the cathode material and the gas filled in in the tube. This showed that electrons are a fundamental component of all atoms in every element. The charge of an electron is -1.602×10^{-19} C, taken as -1.

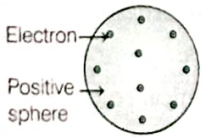
Know the Scientist



J. J. Thomson's most significant discovery was that of the electron, the first subatomic particle to be identified. He received the Nobel Prize in Physics in 1906 for his studies of the electrical conductivity of gases. As head of the famous Cavendish Laboratory in Cambridge, he guided many scientists, including Ernest Rutherford.

Thomson's Model of an Atom (Plum Pudding Model)

- Thomson proposed that an atom is a sphere of positive charge with electrons distributed throughout it, like seeds in a watermelon or plums in a pudding.



This is known as the **Plum Pudding Model** of the atom.

Analogy Imagine a watermelon: the red pulp represents the positively charged matter spread throughout the atom, while the seeds represent the electrons embedded within it.



The Gold Foil Experiment

- In 1911, **Geiger** and **Marsden**, working under Ernest Rutherford, tested Thomson's model through the famous gold foil experiment (also called the **α-ray** scattering experiment).

The Experiment

- A narrow beam of **alpha (α) particles** (tiny, positively charged helium particles emitted from radioactive elements) was allowed to fall at an extremely thin sheet of gold foil. Alpha particles are actually the nuclei of helium atoms, containing two protons and two neutrons.

Observations

- Most alpha particles passed straight through the gold foil without any deflection.
- Some particles were deflected through small angles.
- A very few particles (about 1 in 20,000) bounced back almost in the reverse direction.

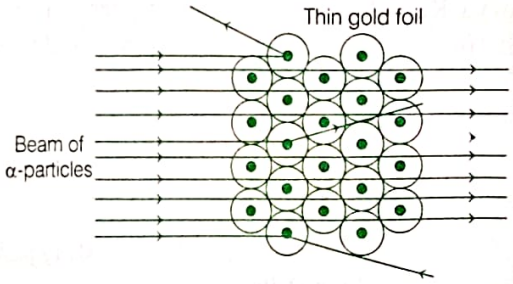


Fig.91 Schemtic view of the gold foil experiment

Conclusion

Thomson's model predicted that all alpha particles should pass straight through or be deflected only slightly (since positive charge was spread evenly). The large deflections and back-scattering of α-rays were totally unexpected, and Thomson's model completely failed to explain these results.

A. Rutherford's Model of an Atom (Planetary Model)

Based on the gold foil experiment, Rutherford proposed a new model

- Most of the space in an atom is empty, which is why most of α-particles passed through gold foil undeflected.

- All the positive charge and most of the mass of an atom is concentrated in an extremely tiny, dense region at the centre of an atom called the **nucleus**.
- Electrons revolve around the nucleus, just like planets orbiting the Sun. This is why it is called the **planetary model of the atom**.

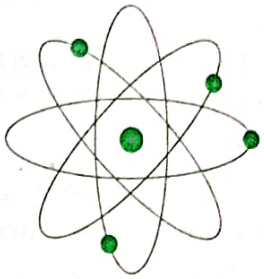


Fig. 9.2 Planetary model suggested by Rutherford

Expert Insight

Size of the Nucleus The nucleus is about 10^5 (one lakh) times smaller than the atom. The diameter of an atom is about 10^{-10} m, while the nucleus is only about 10^{-15} m in diameter. If an atom were the size of a cricket ground (about 100 m), the nucleus would be just the size of a tiny black pepper grain!

B. Limitations of Rutherford's Model

- Rutherford proposed a central nucleus, which was a major advancement. However, his model could not explain atomic stability.
- Electrons move in circular paths, so they are continuously accelerating.
- According to science, accelerating charge emit energy (radiation). Losing energy, electrons should spiral into the nucleus. This would cause the atom to collapse. But atoms are stable in reality, creating contradictions. Hence, Rutherford's model was incomplete.

C. Discovery of the Proton

- Rutherford further showed that the nucleus carries positive charges due to particles called **protons**. Protons are much heavier than electrons and possess a charge equal in magnitude but opposite in sign (+1) to that of electrons (−1).
- For an atom to be electrically neutral, the number of protons must be equal to the number of electrons.

Bohr's Model of the Atom

- To solve the problem of atomic stability, Niels Bohr proposed a new model in 1913. According to Bohr's model
- Electrons do not move randomly around the nucleus. They follow fixed circular paths called stationary states, orbits, or shells. Each shell has a definite amount of energy, so they are also called **energy levels**.
- These shells are represented by the letters $K, L, M, N, \ldots$ or by the numbers $n = 1, 2, 3, 4, \ldots$ The K-shell ($n = 1$) is closest to the nucleus and has the least energy. Energy increases as shells move away from the nucleus.
- Electrons can revolve only in these allowed shells and do not lose energy while revolving in a fixed shell.
- An electron can move to another shell only by absorbing or releasing a fixed amount of energy equal to the difference in energy between the two shells.
- Each shell can hold only a certain maximum number of electrons.

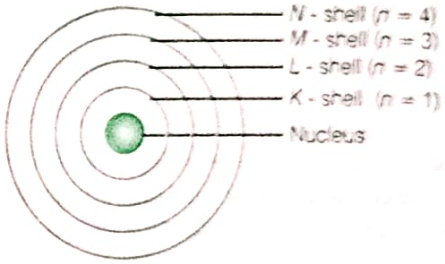


Fig. 9.3 Energy levels in an atom

Know the Scientist



- Born in New Zealand, **Ernest Rutherford** moved to Cambridge to work with J. J. Thomson. He later became known as the **'Father of Nuclear Physics'**. He discovered the atomic nucleus and explained how some elements naturally break down (radioactivity), for which he won the 1908 Nobel Prize in Chemistry.



- **Niels Bohr** was a professor of physics at Copenhagen University, Denmark. He was curious about how atoms remain stable. His atomic model provided a much clearer explanation of the atomic structure. He received the Nobel Prize in Physics in 1922 for his work on the structure of the atom.

What Components Contributes to the Mass of an Atom?

- Rutherford's model showed that most of an atom mass is in its nucleus, while electrons are so light that their mass can be ignored.
- However, a puzzle remained, a hydrogen atom has one proton, while a helium atom has two protons, yet helium is about four times heavier than hydrogen, not just twice.

- Something in the nucleus must be adding mass without adding charge.
- These are the components which contribute to the total mass of an atom.

Discovery of the Neutron

- In 1932, **James Chadwick** (a student of Rutherford) discovered a new subatomic particle with a mass nearly equal to that of a proton but with no electrical charge.
- This neutral particle was named the **neutron (symbol n^0)**. Neutrons are found in the nucleus of all atoms except hydrogen.

Know the Scientist



James Chadwick, working under Rutherford at the Cavendish Laboratory, Cambridge, discovered the neutron in 1932. This breakthrough explained why atomic masses are larger than the sum of just proton masses. He was awarded the Nobel Prize in Physics in 1935. His discovery opened a new era in nuclear physics, enabling the creation of artificial radioactive elements and the splitting of atoms.

The mass of an atom comes mainly from its protons and neutrons packed tightly in the nucleus. This is why atoms are heavier than the mass of their protons alone.

Symbols and Relative Charge of Subatomic Particles

Subatomic Particle	Symbol	Relative Charge	Location
Electron	e^-	−1	Around nucleus
Proton	p^+	+1	In nucleus
Neutron	n^0	0	In nucleus

Curiosity Corner

Lighter atoms often have equal numbers of protons and neutrons (e.g. carbon has 6 protons and 6 neutrons; oxygen has 8 each). As atoms get heavier, the nuclei have many more neutrons than protons (iron has 26 protons and 30 neutrons; uranium has 92 protons and 146 neutrons). Neutrons help reduce repulsion between protons and strengthen the nuclear force that binds the nucleus together.

Symbols of Elements

To make chemistry easier, scientists use short symbols to represent elements. John Dalton introduced the first pictorial symbols in 1803 to represent the known elements.

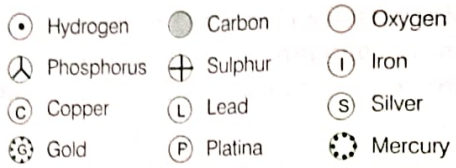


Fig. 9.4 Symbols of some elements given by Dalton

In 1813, Berzelius suggested using letters from the Latin names of elements. Today, the International Union of Pure and Applied Chemistry (IUPAC) approves the names and symbols of all elements.

Rules for Writing Chemical Symbols

- Many symbols use the first letter or the first two letters of the element's English or Latin name.
- The first letter of the symbol is always a capital (uppercase), while the second letter (if present) is always lowercase. e.g. Hydrogen = H; Aluminium = Al (not AL); Cobalt = Co (not CO).
- Some symbols are formed from the first letter and a letter other than the second.
  e.g. Chlorine = Cl; Zinc = Zn.
- Symbols of some symbols come from Latin, Greek, or German names. e.g., Iron = Fe (from Latin ferrum); Mercury = Hg (from Greek hydrargyros); Tungsten = W (from German wolfram).

Name of some Common Elements and their Symbol

Element	Symbol	Element	Symbol	Element	Symbol
Aluminium	Al	Copper (*Cuprum*)	Cu	Nitrogen	N
Argon	Ar	Fluorine	F	Oxygen	O
Barium	Ba	Gold (*Aurum*)	Au	Potassium (Kalium)	K
Boron	B	Hydrogen	H	Silicon	Si
Bromine	Br	Iodine	I	Silver (Argentum)	Ag
Calcium	Ca	Iron (*Ferrum*)	Fe	Sodium (Natrium)	Na
Carbon	C	Lead (*Plumbum*)	Pb	Sulphur	S
Chlorine	Cl	Magnesium	Mg	Uranium	U
Cobalt	Co	Neon	Ne	Zinc	Zn

Note Scientists use symbols instead of full names because they are internationally recognised and allow scientists worldwide to communicate clearly, regardless of language.

Atomic Number

- The atoms of one element differ from those of another in the number of protons (and electrons) they contain. The number of protons in the nucleus of an atom of an element is known as its **atomic number**, designated by the symbol **Z**. The atomic number identifies an element.

Atomic Number (Z) = Number of protons in the nucleus

- Since an atom is electrically neutral, the number of protons must be equals the number of electrons. For example, hydrogen has 1 proton and 1 electron ($Z = 1$). Helium has 2 protons and 2 electrons ($Z = 2$).

Example 01 **What is the atomic number of sodium, which has 11 protons?**

Sol. Atomic number of element = Protons in that element.
Atomic number $Z = 11$. Since the atom is neutral, it also has 11 electrons.

Mass Number

- The total number of protons and neutrons present in the nucleus of an atom is called its mass number, denoted by A. Protons and neutrons together are called **nucleons**.

Mass Number (A) = Number of protons + Number of neutrons

- In the standard notation, an element X with atomic number Z and mass number A is written as

$$\begin{matrix} \text{Mass Number} \\ \text{Atomic Number} \end{matrix} \boxed{\text{Symbol of element}}$$

- For example, the symbol for carbon is C, its atomic number is 6, and its mass number is 12. In notation, it would be written as ${}^{12}_{6}C$.

Note The electron has almost negligible mass compared to a proton or neutron. Therefore, the mass number (A) is the mass of the nucleons (protons + neutrons).

Mass Number of Different Elements

Element	Protons (p^+)	Neutrons (n^0)	Mass Number (A)
Hydrogen	1	0	1
Helium	2	2	4
Lithium	3	4	7
Carbon	6	6	12
Nitrogen	7	7	14
Oxygen	8	8	16
Sodium	11	12	23
Chlorine	17	18	35

Example 02 **An atom has 11 protons and 12 neutrons. Find its (i) atomic number and (ii) mass number.**

Sol. (i) Atomic number (Z) = 11;
(ii) Mass number (A) = 11 + 12 = 23. This is sodium (Na).

Example 03 **An atom has atomic number 17 and mass number 35. Find the number of neutrons.**

Sol. Atomic number (Z) = 17
Mass number (A) = 35
Number of neutrons = $A - Z = 35 - 17 = 18$. This is chlorine (Cl).

Example 04 **An atom has 26 protons and 56 nucleons. Find the number of electrons and neutrons.**

Sol. Atomic number $Z = 26 \rightarrow$ Electrons = 26;
Neutrons = 56 − 26 = 30. This is iron (Fe).

How are Electrons Distributed in Different Energy Levels?

Bohr and Bury suggested the following rules for filling electrons into different shells

- The maximum number of electrons in a shell is given by the formula $2n^2$, where n is the shell number.
- K-shell ($n = 1$): maximum $2 \times 1^2 = 2$ electrons
- L-shell ($n = 2$): maximum $2 \times 2^2 = 8$ electrons
- M-shell ($n = 3$): maximum $2 \times 3^2 = 18$ electrons
- The maximum number of electrons in the outermost shell is 8 (except the K-shell, which holds a maximum of 2 electrons).
- Electrons are filled in different shells in order from the one closest to the nucleus to outward $K \rightarrow L \rightarrow M \rightarrow N$ .... A shell must be completely filled before the next one starts filling. Number of electrons and the electron distribution in the shells of the first eighteen elements is given as follows.

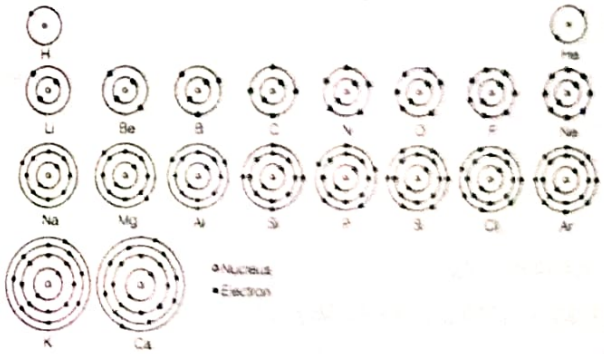


Fig. 9.5 Schematic atomic structure of the first twenty elements

Maximum electrons in a shell = $2n^2$ (n = shell number: 1, 2, 3, ...)

Electronic Configuration

The distribution of electrons among the various shells of an atom is called its electronic configuration. It is written as the number of electrons in each shell, separated by commas (*K*, *L*, *M*, *N*).

Electronic Distribution of the first Eighteen Elements

Element	Symbol	Z	p^+	n^0	e^-	K	L	M
Hydrogen	H	1	1	0	1	1	—	—
Helium	He	2	2	2	2	2	—	—
Lithium	Li	3	3	4	3	2	1	—
Beryllium	Be	4	4	5	4	2	2	—
Boron	B	5	5	6	5	2	3	—
Carbon	C	6	6	6	6	2	4	—
Nitrogen	N	7	7	7	7	2	5	—
Oxygen	O	8	8	8	8	2	6	—
Fluorine	F	9	9	10	9	2	7	—
Neon	Ne	10	10	10	10	2	8	—
Sodium	Na	11	11	12	11	2	8	1
Magnesium	Mg	12	12	12	12	2	8	2
Aluminium	Al	13	13	14	13	2	8	3
Silicon	Si	14	14	14	14	2	8	4
Phosphorus	P	15	15	16	15	2	8	5
Sulphur	S	16	16	16	16	2	8	6
Chlorine	Cl	17	17	18	17	2	8	7
Argon	Ar	18	18	22	18	2	8	8

Combining Capacity of an Atom: Valency

- Atoms combine with other atoms to form molecules and compounds. The number of atoms of hydrogen (or chlorine) with which one atom of an element can combine is called its combining capacity or valency.
- It is expressed in terms of hydrogen and chlorine because both possess a combining capacity of one.

Valence Shell and Valence Electrons

The outermost shell of an atom is called its valence shell. The electrons present in the valence shell are called valence electrons. The number of valence electrons determines how an atom reacts.

The Octet Rule

It has been observed that elements whose outermost shell contains 8 electrons (octet) or 2 electrons in the case of helium - are highly stable and unreactive. Atoms with incomplete valence shells tend to lose, gain, or share electrons to attain a complete octet.

Determining Valency

The number of electrons gained, lost or shared to complete an octet is called the valency of an element.

- If an atom has fewer than 4 valence electrons, it tends to **lose** them to achieve stability (form positive ions).
- If an atom has more than 4 valence electrons, it tends to **gain** electrons to complete its octet (form negative ions).
- If an atom has exactly 4 valence electrons, it tends to **share** electrons (form covalent bonds).
- If an atom has 8 valence electrons (complete octet), it is stable and its valency is 0 (e.g. noble gases like neon and argon).

Example of Valency

- **Sodium (Na)** has electronic configuration 2, 8, 1. It has 1 valence electron. It loses this electron to get an octet (2, 8). Thus, valency = 1.
- **Oxygen** (O) has electronic configuration 2, 6. It has 6 valence electrons. It gains 2 electrons to complete its octet (2, 8). Valency = 2.
- **Carbon** (C) Electronic configuration 2, 4. It has 4 valence electrons and tends to share all four. Valency = 4.
- **Neon** (Ne) Electronic configuration 2, 8. It already has a complete octet. It neither gain nor lose any electron. Valency = 0.

Isotopes

Dalton proposed that all atoms of an element are identical. Later, scientists discovered that atoms of the same element have the same number of protons (same atomic number) but different numbers of neutrons (different mass numbers). Such atoms are called **isotopes.**

Isotopes: Same *Z* (atomic number) but different *A* (mass number)

Isotopes of Hydrogen

Naturally occurring hydrogen is a mixture of three isotopes

- $^{1}_{1}H$ (Protium) – 1 proton, 0 neutrons (~99.98% in nature)

- $^{2}_{1}H$ (Deuterium) – 1 proton, 1 neutron (~0.015%)
- $^{3}_{1}H$ (Tritium) – 1 proton, 2 neutrons (trace amounts)

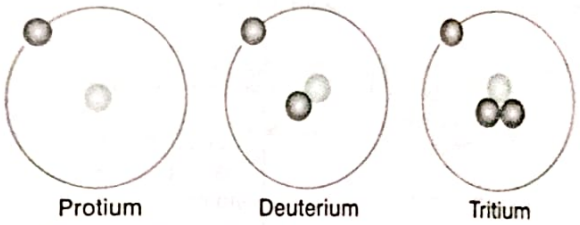


Fig. 9.6 Shematic representation of isotopes of hydrogen

Isotopes of Carbon

Carbon has three isotopes: $^{12}_{6}C$ (most abundant), $^{13}_{6}C$, and $^{14}_{6}C$. All have 6 protons and 6 electrons but differ in the number of neutrons.

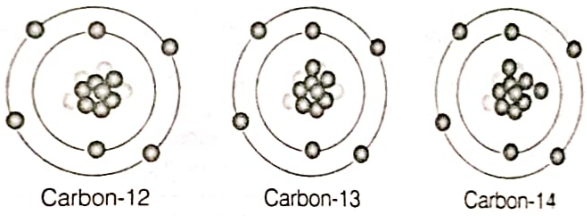


Fig. 9.7 Schematic representation of isotopes of carbon

Chemical Properties of Isotopes Isotopes of an element have similar chemical properties because they have the same number of electrons and the same electronic configuration. However, they differ in their physical properties (e.g. boiling point, melting point) because they have different masses.

Applications of Isotopes

Some isotopes have special properties that are useful in various fields. Some of the applications of isotopes are

- $^{235}_{92}U$ **(uranium-235)** used as a fuel in nuclear reactors to generate electricity.
- $^{60}_{27}Co$ **(cobalt-60)** radioactive isotope used in radiation therapy for cancer treatment.
- $^{131}_{53}I$ **(iodine-131)** used to treat goitre and thyroid cancer.
- $^{14}_{6}C$ **(carbon-14)** used in archaeology and geology to determine the age of ancient fossils and artefacts (carbon dating).

A. Average Atomic Mass

- Most elements exist as a mixture of isotopes, the atomic mass of an element is calculated as the weighted average atomic mass accounting for the relative abundance of each isotope in nature.
- No single chlorine atom has a mass of 35.5 u. It means that in a large sample of chlorine atoms, the weighted average mass found to be 35.5 u, accounting for the abundance of each isotope ^{35}Cl (75%) change and ^{37}Cl (25%).

Mathematically, $\left(35\times\frac{75}{100}+37\times\frac{25}{100}\right)=\frac{142}{4}\text{ u}=35.5\text{ u}$

Know the Scientist



Homi Jehangir Bhabha was an Indian physicist known as the father of the Indian nuclear programme. He made pioneering contributions to atomic energy in India and established key institutions like the Tata Institute of Fundamental Research (TIFR) and the Bhabha Atomic Research Centre (BARC), for peaceful uses of atomic energy including electricity generation, agriculture, and advanced medical treatments.

Isobars

- When atoms of different elements have the same mass number, but different atomic number, they are called isobars.

Isobars : Different Z (atomic number) but same A (mass number)

- For example, calcium ($Z = 20$), potassium ($Z = 19$), and argon ($Z = 18$) all have a mass number of 40. They are different elements but are isobars of each other because they all have 40 nucleons in their nuclei.

Comparision of Isotopes and Isobars

Property	Isotopes	Isobars
Atomic number (Z)	Same	Different
Mass number (A)	Different	Same
Chemical properties	Similar	Different
Example	$^{1}_{1}H$, $^{2}_{1}H$, $^{3}_{1}H$ (hydrogen)	$^{40}_{18}Ar$, $^{40}_{19}K$, $^{40}_{20}Ca$

The journey of exploring the mysteries of the atom is far from over, and exciting discoveries still lie ahead.

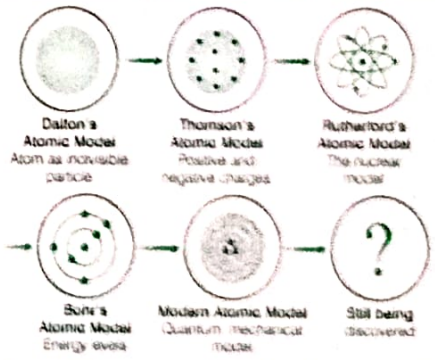


Fig. 9.8 Journey of the development of atomic models.

Mind MAP The Ultimate Revision Tool for Exams

JOURNEY INSIDE THE ATOM

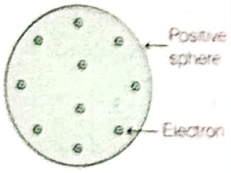
Bohr's Model
The main postulates of this model are
- Electrons revolve around the nucleus in a circular orbit.
- Only certain special orbits called **discrete orbits** or **energy levels** of electrons are allowed inside the atom.
- While revolving in discrete orbits, the electrons do not radiate energy.
- Energy levels or shells are designated as K, L, M, N etc., or by the numbers, n = 1, 2, 3, 4, etc.



A few energy levels in an atom

Atomic Models
Explain the arrangement of sub-atomic particles within an atom.

Thomson's Model
It considers the atom to be like a **plum pudding** in which the electrons were embedded in a sphere of positive charge like currents in a pudding. The negative and positive charges balance each other thus atom as a whole is electrically neutral.



Limitations
It failed to explain the results of α-particles scattering experiment of Rutherford.

Rutherford's Model
On the basis of the α-particles scattering experiment, Rutherford stated that
- There is a positively charged, highly densed centre in an atom, called the nucleus.
- The nucleus is very small as compared to the size of the atom.
- Nearly the whole mass of an atom resides in it.
- The electrons revolve around the nucleus in circular paths.

Limitations
It failed to explain the stability of atoms.

Characteristics of Atom
These are the properties by which an atom can be identified.

Atomic Number (Z)
- It is equal to the number of protons present in the nucleus of an atom.
- It is denoted by Z.

Mass Number (A)
It is the sum of the number of protons and neutrons present in the nucleus of an atom.
$A = p + n$

Different Type of Atomic Species

Sub-Atomic Particles
These are the fundamental particles from which an atom is made.

Proton
- It was discovered by E. Goldstein in 1886.
- It is represented as p^+. These are the positively charged particles.
- Its mass was approximately 2000 times as that of the electron.
- Its mass is taken as one unit and charge as +1.

Anode or Canal Rays are the beam of protons but do not originate from anode.

Neutrons
These were discovered by J. Chadwick in 1932.
In general, a neutron is represented as 'n'.
These are electrically neutral particles.
These are present inside the nucleus along with the protons.

Electron
- It was identified by J.J. Thomson.
- It is represented as 'e^-'.
- It is the negatively charged particle present in an atom.
- Its charge is taken as –1. The mass of an electron is considered to be negligible.

Distribution of Electrons in Different Orbits
- Suggested by Bohr and Bury.
- Maximum number of electrons present in a shell = $2n^2$ (Here, n = 1,2,3...)
- Outermost orbit can accommodate a maximum of 8 electrons.
- The shells are filled in a step-wise manner.

Valence Electrons
It is the number of electrons present in the outermost or valence shell of an atom.

Valency
It is the combining capacity of an element with the atom(s) of other element(s) in order to complete its octet.

Isotopes
- The atoms of the same element, having same atomic number but different mass number.
  e.g. , ${}_1H^1$, ${}_1H^2$, ${}_1H^3$
- Their chemical properties are same due to same number of valence electrons.
- Their physical properties are different because of having different atomic masses.

Applications
- An uranium isotopes is used as a fuel.
- A cobalt isotopes is used in the treatment of cancer.
- An iodine isotope is used in the treatment of goitre.

Isobars
- Atoms of different elements with different atomic numbers but same mass number.
  e.g. ${}^{40}_{18}Ar$ and ${}^{40}_{20}Ca$.
- Their chemical properties are different because of different atomic number.

NCERT Folder

Answers to all Questions of NCERT Textbook 'Exploration'

Pause and Ponder

Q.1 Suppose you made up your own 'atom', as Thomson described, using clay for the positive charge and small beads for the electrons spread through it. What will happen if

(i) the positive charge on the clay is lesser than the total negative charge of the beads?

(ii) by mistake, the clay itself carries a bit of negative charge? Would your model still represent a neutral atom?

Sol. (i) If the positive charge of the clay is less than the total negative charge of the beads, the extra negative charge will increase on atom. As a result, the entire atom will become negatively charged instead of being neutral.

(ii) If the clay itself carries a bit of negative charge along with the positive charge, it will add up with the negative charge of the beads. This means the total negative charge will become greater than the positive charge. Therefore, the model will no longer represent a neutral atom.

Q.2 Could an orange or a lemon, which also contain seeds inside soft pulp, be a good comparison? In what ways does it match Thomson's idea and where does it fall short?

Sol. Yes, an orange or a lemon can be a fairly good comparison to Thomson's model. It matches the idea correctly because the soft pulp is spread everywhere like the positive charge, and the seeds are spread inside just like the negatively charged electrons.

However, it falls short because the seeds in a lemon are mostly gathered near the center, whereas Thomson suggested that electrons are spread uniformly everywhere. Also, this fruit comparison does not show the true empty space that actually exists inside a real atom.

Q.3 Why did Thomson conclude that electrons are present in all atoms?

Sol. Thomson concluded this because, during his famous cathode ray tube experiment, he noticed that the nature of the cathode rays remained exactly the same even when he changed the gas inside the tube or used different materials for the electrodes. Since these tiny negatively charged particles (electrons) appeared exactly identical no matter what material was used, he then realised that they must be a basic, universal part of all atoms.

Q.4 What do you think would happen if α-particles were replaced with negatively charged particles in Rutherford's gold foil experiment?

Sol. If negatively charged particles, like electrons, were used instead of the positively charged alpha particles, they would behave very differently. Because opposite charges attract, these negative particles would be strongly attracted to the positive nucleus instead of being pushed away or deflected. Furthermore, since electrons are incredibly light compared to heavy alpha particles, they would easily lose their path and get captured by the atoms, completely changing the results of the experiment.

Q.5 Rutherford found that a few α-particles bounced back sharply. How does this single surprising result completely rule out Thomson's 'plum pudding model' of the atom?

Sol. According to Thomson's plum pudding model, the positive charge and mass were spread out evenly and very thinly like a soft cloud. If this were true, the heavy and fast alpha particles should have easily passed straight through the gold foil.

The fact that a few alpha particles hit something hard and bounced straight back proved that all the positive charge and heavy mass must be packed tightly into a tiny, dense center. This dense center (the nucleus) is the reason which hit α-rays back so hard, which completely disproved Thomson's idea of a plum pudding model.

Q.6 If you could ask Rutherford one question about his work, what would it be?

Sol. If I could ask Rutherford one question, I would ask: "When you saw those alpha particles bounce straight back at you like a bullet hitting tissue paper, what was the very first thought that came to your mind before you discovered it was a dense nucleus?"

Q.7 **Assertion** (A) Rutherford concluded that most of the mass of an atom is concentrated in a small region at the centre called the nucleus.

Reason (R) According to Thomson's model, electrons are embedded in a uniformly distributed positive charge sphere.

Choose the correct option.

(i) Both A and R are true, and R is the correct explanation of A.

(ii) Both A and R are true but R is not the correct explanation of A.

(iii) A is true, but R is false.

(iv) A is false, but R is true.

Sol. The correct option is (ii) Both A and R are true, but R is not the correct explanation of A.

Rutherford discovered the nucleus, and Thomson proposed the uniform positive sphere. However, Rutherford discovered the heavy central mass because heavy alpha particles bounced back during his gold foil experiment, not because of what Thomson's earlier model had stated.

Q.8 Imagine you are a scientist who has discovered a new element. Name this element after yourself and justify that the symbol you have chosen follows the IUPAC rules.

Sol. Let us imagine my name is "Ayan," and I proudly name my newly discovered element "Ayanium." According to the IUPAC rules, I will choose the symbol "Ay". To justify this, the first letter 'A' is written in capital (uppercase), and the second letter 'y' is written in small (lowercase), following the same uppercase-lowercase rule, which makes it scientifically correct.

Q.9 What problems could arise if every scientist used different symbols for the same element?

Sol. If every scientist used different symbols for the same element, it would create a lot of confusion across the world. Scientists from different countries would not be able to read or understand each other's research, chemical formulas, or discoveries.

Sharing scientific knowledge and studying chemistry together would become extremely difficult because there would be no common language to unite them.

Q.10 An atom with an atomic number of 26 has 56 nucleons. Find out its number of electrons, protons and neutrons.

Sol. The atomic number of an element is the same as the number of protons in that element. So, it has 26 protons.

Since a normal atom is neutral, the number of electrons is equal to the number of protons. Thus, it has 26 electrons.

The term "nucleons" refers to the total number of protons and neutrons together (Mass Number = 56).

To find the neutrons, we subtract the protons from the mass number: $56 - 26 = 30$. Therefore, the atom has 30 neutrons.

Q.11 The nucleus of an atom contains 20 protons. If its mass number is 41, find the number of neutrons in it.

Sol. The mass number of an atom is the sum of its protons and neutrons. It is given that the mass number is 41 and the number of protons is 20.

To find the number of neutrons, we subtract the protons from the mass number: $41 - 20 = 21$. Therefore, the atom contains 21 neutrons.

Q.12 An atom has 18 neutrons and an atomic number of 17. What is its mass number?

Sol. The atomic number tells exactly how many protons any atom has. Here, the atomic number is 17, which means there are 17 protons.

The mass number is calculated by adding the number of protons and the number of neutrons together.

So, Mass Number = 17 + 18 = 35.

Therefore, the mass number of this atom is 35.

Q.13 An atom ^{24}A has 11 electrons. Find the number of neutrons in it.

Sol. The general symbol of an element is $^{A}_{Z}A$. For any neutral atom, no. of protons = no. of electrons.

So, the number of protons is 11, and the mass number is 24.

To find the number of neutrons, we subtract the number of protons from the mass number:

$24 - 11 = 13$

Therefore, there are 13 neutrons in this atom.

Q.14 Identify the number of electrons in the outermost shell of the following elements

(i) ${}^{12}_{6}C$ (ii) ${}^{19}_{9}F$ (iii) ${}^{28}_{14}Si$

Sol. (i) **Carbon** (${}^{12}_{6}C$) The atomic number (bottom number) is 6, meaning it has 6 electrons. Its electronic configuration is 2, 4. Therefore, it has 4 electrons in its outermost shell.

(ii) **Fluorine** (${}^{19}_{9}F$) The atomic number is 9, meaning it has 9 electrons. Its electronic con-figuration is 2, 7. Therefore, it has 7 electrons in its outermost shell.

(iii) **Silicon** (${}^{28}_{14}Si$) The atomic number is 14, meaning it has 14 electrons. Its electronic configuration is 2, 8, 4. Therefore, it has 4 electrons in its outermost shell.

Q.15 Write the electronic configuration of the elements having atomic numbers 12, 16 and 18.

Sol. To write the electronic configurations, we fill the shells in the order of $K(2)$, $L(8)$ and $M(18)$.

- **For atomic number 12 (Magnesium)** The electrons are distributed as 2, 8, 2.
- **For atomic number 16 (Sulphur)** The electrons are distributed as 2, 8, 6.
- **For atomic number 18 (Argon)** The electrons are distributed as 2, 8, 8.

Q.16 **Solve this riddle** I am an atom with a mass number of 23 and 11 protons. I am a soft metal and react vigorously with water. Who am I and how many neutrons do I have? You can also create one such riddle.

Sol. Since you have exactly 11 protons, your atomic number is 11. The element with atomic number 11 is Sodium (Na), which is indeed a soft metal that reacts strongly with water. To find neutrons, I will subtract the protons from the mass number: $23 - 11 = 12$. You have 12 neutrons.

My Riddle I am an atom with 8 protons and a mass number of 16. I am an invisible gas that all living beings breathe in to stay alive. Who am i and what is my electronic configuration?

(Answer : Oxygen and its electronic configuration is 2, 6).

Q.17 Two different atoms have 11 protons each, but one has 12 neutrons, and the other has 13 neutrons. How do their atomic numbers and mass numbers compare? Are they the same element or different elements?

Sol. Both atoms contain exactly 11 protons, their atomic numbers are identical (both have $Z = 11$).

However, their mass numbers will be different. The first atom has a mass number of 23 (11 + 12), while the second atom has a mass number of 24 (11 + 13).

Since their atomic number is exactly the same, they belong to the same element (which is sodium). These different forms of the same element, having different masses, are known as isotopes.

Q.18 If a bromine atom is available in the form of, say two isotopes, ${}^{79}_{35}Br$ (49.7%) and ${}^{81}_{35}Br$ (50.3%), calculate the average atomic mass of the bromine atom.

Sol. The average atomic mass is calculated by multiplying the mass of each isotope by its percentage of occurrence in nature, and then adding those values together.

Average atomic mass

$$= \left(79 \times \frac{49.7}{100}\right) + \left(81 \times \frac{50.3}{100}\right)$$

$$= (79 \times 0.497) + (81 \times 0.503)$$

$$= 39.263 + 40.743 = 80.006 \text{ u}$$

Therefore, the average atomic mass of the bromine atom is approximately 80.006 u.

Revise, Reflect, Refine

Q.1 Choose the correct options and explain the reason for the correct and incorrect options in the context of Ernest Rutherford's gold foil experiment.

(i) The experiment clearly showed the existence of neutrons in the nucleus.

(ii) The results disproved the plum pudding model and led to the idea of a nucleus at the centre of the atom.

(iii) The large deflection of a few alpha particles indicated that most of the mass of the atom and positive charge are packed into a tiny centre.

(iv) The way alpha particles were deflected showed that electrons move around the nucleus.

Sol. The correct options are (ii) and (iii).

Reason for correct options During the experiment, a few heavy alpha particles bounced back sharply.

This proved that the atom's positive charge and mass are not spread out evenly (which disproves Thomson's plum pudding model). Instead, they are tightly packed into a tiny, dense centre called the nucleus.

Reason for incorrect options Option (i) is incorrect because neutrons were discovered much later by James Chadwick, not by Rutherford's experiment. Option (iv) is incorrect because the deflection of alpha particles was caused by the heavy, positive nucleus, not by the very light electrons.

Q.2 Which of the following statements are correct or incorrect according to the Bohr's atomic model? Give a reason for each statement.

(i) Electrons lose energy while moving in fixed orbits and slowly fall into the nucleus.

(ii) Electrons can exist anywhere around the nucleus with no fixed energy.

(iii) Electrons revolve around the nucleus in orbits of fixed energy without losing energy.

(iv) Electrons can be found between energy levels as they move around the nucleus.

Sol. (i) **is incorrect.** According to Bohr, electrons revolve in special stationary orbits and do not lose energy while moving in them.

(ii) **is incorrect.** Electrons cannot exist just anywhere. They must revolve only in specific, allowed circular paths called energy shells.

(iii) **is correct.** This is the main principle of Bohr's model. Electrons revolve in fixed energy shells (like *K, L, M*) and their energy remains constant as long as they stay in that shell.

(iv) **is incorrect.** Electrons cannot stay in the empty spaces between the energy levels. They can jump from one level to another, but cannot be found in between.

Q.3 The composition of the nuclei of three atomic species *X*, *Y*, and *Z* are given as follows

	X	*Y*	*Z*
Number of protons	18	17	17
Number of neutrons	19	18	20

Explain the relation between the following.

(i) *Y* and *Z* (ii) *Z* and *X*.

Sol. (Based on the data provided for these species: *X* has 18*p*, 19*n*; *Y* has 17*p*, 18*n*; *Z* has 17*p*, 20*n*).

(i) **Relation between *Y* and *Z*** Both *Y* and *Z* have exactly the same number of protons (17), meaning they belong to the same element (Chlorine). However, they have a different number of neutrons (18 and 20). Therefore, *Y* and *Z* are isotopes.

(ii) **Relation between *Z* and *X*** If we calculate their mass numbers (protons + neutrons), *X* has a mass number of 37 (18 + 19), and *Z* also has a mass number of 37 (17 + 20). Since they have the same mass number but different atomic numbers (different protons), Z and X are isobars.

Q.4 What conclusion did Rutherford draw about the position and characteristics of the atom's positively charged part based on the few alpha particles that bounced back or were deflected at large angles in the gold foil experiment?

Sol. According to Rutherford's experiment, a few fast and heavy alpha particles bounced almost straight back, Rutherford concluded that the positive charge of an atom is not spread out like a soft cloud. Instead, he realised that all the positive charge and almost the entire heavy mass of the atom are tightly packed into a very tiny, hard, and dense region exactly at the centre of the atom. He named this central part the nucleus.

Q.5 Explain and arrange the following statements in the correct chronological order to show how atomic models have evolved over time.

(i) Bohr's model proposed that electrons move in fixed orbits around the nucleus, each with a definite energy.

(ii) Thomson's model depicted the atom as a 'plum pudding' with electrons embedded in a sphere of positive charge.

(iii) Rutherford's model proposed that atoms have a dense central nucleus.

(iv) Dalton's model described atoms as indivisible particles.

Sol. The correct chronological order (from oldest to newest) is: (iv) → (ii) → (iii) → (i).

Explanation First, Dalton proposed that atoms are the smallest, indivisible particles i.e. (iv). Years later, Thomson discovered electrons and gave the 'plum pudding model' to show charges i.e. (ii). After that,

Rutherford conducted his gold foil experiment and discovered the central nucleus. i.e. (iii). Finally, to fix the stability problems in Rutherford's model, Bohr introduced the concept of fixed energy shells i.e. (i).

Q.6 Electrons move around the nucleus in orbits. Why do they not fly away from the atom? Explain what keeps them attracted to the nucleus.

Sol. Electrons do not fly away from the atom because they are held tightly by a strong electrostatic force of attraction. The nucleus contains protons, which are positively charged, while electrons are negatively charged. Since opposite charges naturally attract each other, this strong inward pull from the positive nucleus keeps the fast-moving negative electrons perfectly in their orbits.

Q.7 **Assertion** (A) The discovery of subatomic particles helped in understanding the atomic structure.

Reason (R) The number of electrons is equal to the number of protons in an atom.

Choose the correct option.

(i) Both A and R are true, and R is the correct explanation of A.

(ii) Both A and R are true, but R is not the correct explanation of A.

(iii) A is true, but R is false.

(iv) A is false, but R is true.

Sol. The correct option is (ii) Both A and R are true, but R is not the correct explanation of A.

Discovering electrons, protons and neutrons helped scientists build better atomic models over time.

Also, it is true that a neutral atom has equal electrons and protons. However, the equal number of charges (Reason) is not the actual explanation for how discovering these particles helped to understand the atomic structure (Assertion).

Q.8 Magnesium is essential for many biological processes, including muscle contraction. For an atom of magnesium with a mass number of 24 and atomic number 12, determine the number of (i) protons, (ii) neutrons, (iii) electrons, and also illustrate the arrangement of electrons in a magnesium atom.

Sol. Given for Magnesium: Atomic Number = 12, Mass Number = 24.

(i) **Protons** The atomic number is always equal to the number of protons. Thus, it has 12 protons.

(ii) **Neutrons** Neutrons = Mass Number – Protons. So, 24 – 12 = 12 neutrons.

(iii) **Electrons** In a neutral atom, the number of electrons equals the number of protons. Thus, it has 12 electrons.

Arrangement of electrons The 12 electrons will fill the shells step-by-step. The first *K*-shell takes 2 electrons, the *L*-shell takes 8 electrons, and the remaining 2 electrons go to the *M*-shell. So, the electronic configuration is 2, 8, 2.

Q.9 Find the following information for the elements shown in figure given below.

(i) Name of the element

(ii) Symbol

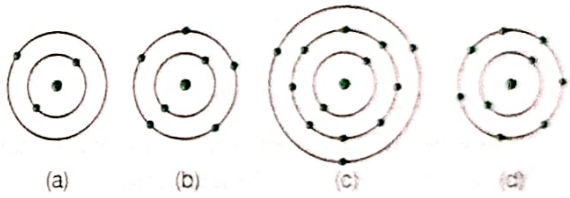
(iii) Total number of electrons

(iv) Number of valence electrons

(v) Valency of the element

(vi) Number of protons

(vii) Atomic number



Sol. (a) • **Element** : Lithium • **Symbol** : Li
• **Total electrons** : 3 • **Valence electrons** : 1
• **Valency** : 1 • **Protons** : 3
• **Atomic number** : 3

(b) • **Element** : Nitrogen • **Symbol** : N
• **Total electrons** : 7 • **Valence electrons** : 5
• **Valency** : 3 • **Protons** : 7
• **Atomic number** : 7

(c) • **Element** : Aluminium • **Symbol** : Al
• **Total electrons** : 13 • **Valence electrons** : 3
• **Valency** : 3 • **Protons** : 13
• **Atomic number** : 13

(d) • **Element** : Fluorine • **Symbol** : F
• **Total electrons** : 9 • **Valence electrons** : 7
• **Valency** : 1 • **Protons** : 9
• **Atomic number** : 9

Q.10 Both Rutherford's and Bohr's models have electrons orbiting the nucleus. Why did Rutherford's model fail to explain atomic stability, while Bohr's model succeeded?

Sol. Rutherford's model failed because, any charged particle moving in a circular path should constantly lose energy in the form of radiation. If this happened, the electron would lose its energy, spiral inward, and crash into the nucleus, making the atom ustable. Bohr's model succeeded because he introduced a new rule: electrons revolve in special "stationary orbits." He stated that as long as an electron stays in its fixed orbit, it simply does not radiate or lose any energy, which perfectly explains why the atom is stable.

Q.11 An atom ^{70}X has 31 electrons. How many neutrons are there in its nucleus?

Sol. A normal atom is electrically neutral, so the number of protons exactly equals to the number of electrons. So, the atom has 31 protons.

Mass Number = Protons + Neutrons.

To find neutrons, we reverse the formula Neutrons = Mass Number – Protons. Neutrons = 70 – 31 = 39.

Thus, there are 39 neutrons in its nucleus.

Q.12 An atom has 79 protons and a mass number of 197. Calculate (i) the number of neutrons, and (ii) the number of electrons.

Sol. (i) **Number of neutrons** Neutrons = Mass Number – Protons. So, 197 – 79 = 118 neutrons.

(ii) **Number of electrons** An atom is electrically neutral, so the number of electrons equals to the number of protons. Thus, it has 79 electrons.

Q.13 Complete the table given below.

Atomic No.	Mass No.	Neutrons	Protons	Electrons	Name of Element
5	—	6	—	—	—
—	14	—	—	7	Nitrogen
—	24	—	12	—	—
15	—	16	—	—	—
—	1	0	—	—	—

Sol. To fill out the table, we use simple atomic rules: Atomic Number = Protons = Electrons; and Mass Number = Protons + Neutrons.

Atomic No.	Mass No.	Neutrons	Protons	Electrons	Name of Element
5	11	6	5	5	Boron
7	14	7	7	7	Nitrogen
12	24	12	12	12	Magnesium
15	31	16	15	15	Phosphorus
1	1	0	1	1	Hydrogen

Q.14 Aman was discussing the structure of atom with his classmates. During the discussion, he learnt that an element X has a mass number of 35 and contains 18 neutrons. Based on this information, answer the following questions.

(i) How many electrons and protons does element X have?

(ii) What is its atomic number?

(iii) Identify the element X.

(iv) Write its electronic configuration.

(v) How many valence electrons does it have?

(vi) What will be the mass number if two neutrons are added to its nucleus?

(vii) What will be the relation of X with the new atom?

Sol. (i) Protons = Mass Number – Neutrons = 35 – 18 = 17. So, it has 17 protons and 17 electrons.

(ii) The atomic number is always equal to the number of protons, so it is 17.

(iii) The element with the atomic number 17 is chlorine (Cl).

(iv) With 17 electrons, shells are filled as $K(2)$, $L(8)$, $M(7)$. The configuration is 2, 8, 7.

(v) Looking at the outermost shell, it has 7 valence electrons.

(vi) If two neutrons are added, the new mass number becomes 35 + 2 = 37.

(vii) Since the number of protons remains the same (17) but the mass number has changed (37), the new atom will be an isotope of element X.

Q.15 In an atom, there are 12 protons and 12 neutrons in the nucleus. Now, imagine that all the electrons are replaced with some hypothetical particles that have the same charge as electrons but are 500 times heavier. What effect will this replacement have on the atom's?

(i) Atomic number

(ii) Atomic mass

(iii) Mass number

(iv) Overall charge

Sol. (i) **Atomic number** It will remain unchanged (12). The atomic number only depends on the number of protons inside the nucleus, which has not changed.

(ii) **Atomic mass** It will increase. Normally, electrons are so light that their weight is ignored. However, these new particles are 500 times heavier, meaning their combined weight will now add a very noticeable amount to the overall mass of the atom.

(iii) **Mass number** It will remain unchanged (24). The mass number is strictly defined as the sum of protons and neutrons in the nucleus (12 + 12), regardless of what is orbiting outside.

(iv) **Overall charge** It will remain unchanged (neutral). Since the new particles carry the exact same negative charge as normal electrons, they perfectly balance the positive charge of the 12 protons just like before.